

## 0/17 Questions Answered

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## hw6-single-prop

Student Name

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**Q1 Problematic interpretation****6 Points**

Cheating has always been a concern in academia, but with the release of ChatGPT in November 2022, there has been increased discussion of cheating and the role that ChatGPT and other AI tools may play.

Consider the following scenario:

Maria, a student in computer science, is conducting a small research study on cheating using ChatGPT.

She obtains a random sample of 20 UCI students and asks them whether they have cheated using ChatGPT. The results show that 10% of the students in her sample have cheated using ChatGPT. She wants to construct a 95% confidence interval for the proportion of people who have cheated using ChatGPT, among all UCI students, so she computes the margin of error, which turns out to be 0.1315.

**Q1.1****2 Points**

What is the upper bound of this CI? Enter only your final answer. Round to four decimal places (e.g. 1.23456 should be rounded to 1.2346). Include a picture of your work.

 Please select file(s)**Select file(s)****Save Answer**

**Q1.2****1 Point**

What is the lower bound of this CI? Enter only your final answer. Round to four decimal places (1.23456 should be rounded to 1.2346). Include a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q1.3****3 Points**

Maria would like to include her results in an article that she is writing for a local newspaper but, when she computes the CI, she feels puzzled: something doesn't seem right about the lower bound. She knows you have taken a stats class, and she comes to you to ask your opinion. What is the issue with the lower bound of her CI? And what is a reason that you might suggest to Maria to explain why she faced this issue?

**Save Answer**

## Q2 Cheating trends

11 Points

AI tools, such as ChatGPT, are powerful tools and we are still learning helpful ways to utilize them. Unfortunately, some think that the introduction of AI has made cheating easier, and therefore more common.

A study was conducted by Lee et al. ([2024](#)) to investigate changes in cheating behavior among US high school students, including the use of AI. They surveyed high school students about their experience with cheating behaviors. One question they asked students is if they had cheated by using AI in the past month. When they sampled 250 public high school students in 2023 they had 38 students report they had cheated using AI in the past month.

### Q2.1

1 Point

First we want to understand what are we investigating and what is our data?

The population parameter of interest for this study is:

Proportion of public high school students surveyed that have cheated using AI in the past month

Proportion of public high school students that have cheated using AI in the past month

Number of public high school students surveyed that have cheated using AI in the past month

Number of public high school students that have cheated using AI in the past month

Save Answer

**Q2.2****1 Point**

The value of the population parameter is:

38

250

$38 / 250 = .152$

$250 - 38 = 212$

unknown

Save Answer

**Q2.3****1 Point**

The point estimate (sample statistic) for this study is:

Proportion of public high school students surveyed that have cheated using AI in the past month

Proportion of public high school students that have cheated using AI in the past month

Number of public high school students surveyed that have cheated using AI in the past month

Number of public high school students that have cheated using AI in the past month

Save Answer

## Q2.4

1 Point

The value of the sample statistic is:

38

250

 $38 / 250 = .152$  $250 - 38 = 212$ 

unknown

[Save Answer](#)

## Q2.5

1 Point

What are the null and alternative hypotheses to test if a majority of public high school students cheating using AI in the past month.

$H_0$ : The proportion of public high school students that have cheated using AI in the past month is more than 50%;  $H_1$ : The proportion of public high school students that have cheated using AI in the past month is less than 50%

$H_0$ : The proportion of public high school students that have cheated using AI in the past month is 50%;  $H_1$ : The proportion of public high school students that have cheated using AI in the past month is less than 50%

$H_0$ : The proportion of public high school students that have cheated using AI in the past month is 50%;  $H_1$ : The proportion of public high school students that have cheated using AI in the past month is not 50%

$H_0$ : The proportion of public high school students that have cheated using AI in the past month is not 50%;  $H_1$ : The proportion of public high school students that have cheated using AI in the past month is 50%

[Save Answer](#)

**Q2.6****1 Point**

Let's check that the conditions are met to do a hypothesis test for a proportion. For each condition specify if it is met and explain your answer.

Condition 1: Sample data are independent?

Yes

No or unknown

Save Answer

**Q2.7****1 Point**

Condition 2: More than 10 success and 10 failures in the sample?  
Explain your answer.

Yes

No

Save Answer

**Q2.8****1 Point**

Condition 3. Sample size is less than 10% of population? Explain your answer.

Yes

No

Save Answer

**Q2.9****1 Point**

We will proceed assuming all conditions for the hypothesis test were met. If a one-sample test for proportion is performed, which of the following outcomes would lead to a rejection of the null hypothesis at a significance level of 0.05?

A p-value of 0.07

A p-value of 0.03

A p-value of 0.50

A p-value of 1.00

Save Answer

**Q2.10****1 Point**

Conducting a hypothesis test for one proportion results in a p-value of  $5.1 \times 10^{-53}$ . Which of the following is a valid conclusion from this p-value?

We accept the null hypothesis and conclude that the population proportion of public high school students that cheated using AI in the past month is 50%.

We reject the null hypothesis and conclude that the population proportion of public high school students that cheated using AI in the past month is 50%.

We reject the null hypothesis and conclude the population proportion of public high school students that cheated using AI in the past month is not 50%.

We fail to reject the null hypothesis and cannot make a conclusion about the population proportion of public high school students that cheated using AI in the past month.

[Save Answer](#)

**Q2.11****1 Point**

In this study they were not actually thinking a majority of students were cheating. Instead they wanted to understand the true proportion of public high students that are cheating using AI. Computation of a 95% confidence interval yields (11%, 20%). What is an appropriate interpretation of this confidence interval?

The true proportion of public high school students that cheated using AI recently is somewhere between 11% and 20%.

We are confident that the true proportion of public high school students that cheated using AI recently is somewhere between 11% and 20%.

We are 95% confident that the true proportion of public high school students that cheated using AI recently is somewhere between 11% and 20%.

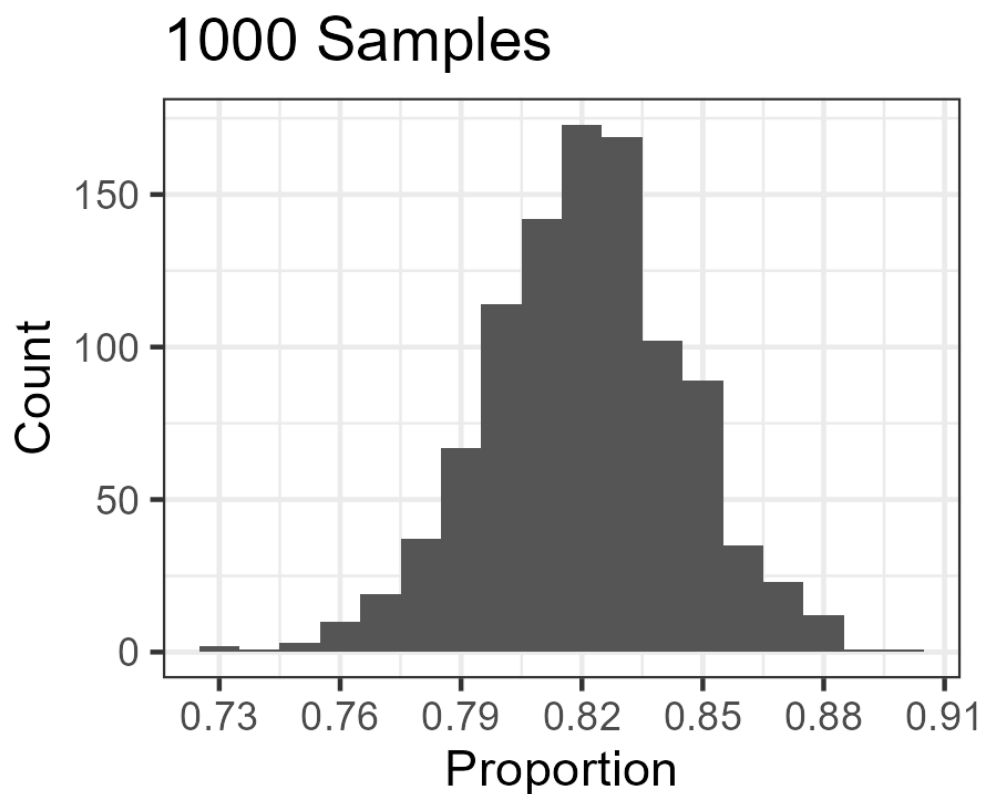
We are confident that there is a 95% probability that the true proportion of public high school students that cheated using AI recently is somewhere between 11% and 20%.

There is a 95% probability that the true proportion of public high school students that cheated using AI recently is somewhere between 11% and 20%.

[Save Answer](#)

**Q3****3 Points**

Say we were able to collect 1000 samples, and for each we calculated the sample proportion, plotted below.

**Q3.1****1 Point**

What is visualized in the plot?

- sampling distribution
- distribution of sample data
- distribution of population data

Save Answer

**Q3.2****2 Points**

Based on this plot, what do you think the value of the true population proportion is? Explain.

[Save Answer](#)**Q4****1 Point**

The following question is related to the data topic we got to learn about, cheating with AI. There are no correct answers, this question is only being graded for completeness (note that answering "It was interesting" is an insufficient response).

Please explain in detail what you liked and/or what you did not like about this data topic.

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